Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two different coils have self-inductance $\mathrm{L_1 = 8mH, L_2 = 2mH}$. The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same rate. At a certain instant of time, the power given to the two coils is the same. as the time the current, the induced voltage and the energy stored in the first coil are $\mathrm{i_1}$ $\mathrm{V_1}$ and W_1 respectively. Corresponding values for the second coil at the same instant are $\mathrm{i_2}$, $\mathrm{V_2}$ and $\mathrm{W_2}$ respectively. Then

(A) $\frac{\mathrm{i_1}}{\mathrm{i_2}} = \frac{1}{4}$

(B) $\frac{\mathrm{i_1}}{\mathrm{i_2}} = \frac{1}{8}$

(C) $\frac{\mathrm{W_2}}{\mathrm{W_1}} = \frac{4}{1}$

(D) $\frac{\mathrm{W_2}}{\mathrm{W_1}} = \frac{8}{1}$

Q.2

A coil of inductance $8.4\mathrm{mH}$ and resistance 6Ω is connected to a $12\ \mathrm{V}$ battery. The current in the coil is $1.0\ \mathrm{A}$ in the time (approx).

(A) $500\mathrm{sec}$

(B) $20\mathrm{sec}$

(C) 35 milli sec

(D) 1 milli sec

Q.3

A coil of inductance $300\mathrm{mH}$ and resistance 2Ω is connected to a sources of voltage $2\ \mathrm{V}$ The current reaches half of its steady state value in

(A) $0.15\ \mathrm{s}$

(B) $0.3\ \mathrm{s}$

(C) $0.05\ \mathrm{s}$

(D) $0.1\ \mathrm{s}$

Q.4

Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $\mathrm{A} = 10\ \mathrm{cm}^2$ and length $= 20\ \mathrm{cm}$. If one of the solenoids has 300 turns and the other 400 turns, their mutual inductance is $\left(\mu_0 = 4\mathrm{p} \times 10^{-7}\ \mathrm{T\ m\ A}^{-1}\right)$

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(A) $4.8\pi \times 10^{-4}\mathrm{H}$

(B) $4.8\pi \times 10^{-5}\mathrm{H}$

(C) $2.4\pi \times 10^{-4}\mathrm{H}$

(D) $2.4\pi \times 10^{-5}\mathrm{H}$

Q.5

The flux linked with a coil at any instant t is given by $\phi = 10t^2 - 50t + 250$ The induced emf at $t = 3$ s is

(A) -190 V

(B) -10 V

(C) 10 V

(D) 190 V

Q.6

A current i ampere flows along an infinitely long straight thin walled tube, then the magnetic induction at any point inside the tube is

(A) $\frac{\mu_0}{4\pi} \cdot \frac{2\mathrm{i}}{\gamma}$ Tesla

(B) zero

(C) infinite

(D) $\frac{2\mathrm{i}}{\gamma}$ Tesla

Q.7

A loop of area $0.1\ \mathrm{m}^2$ rotates with a speed of $60\mathrm{rev}/\mathrm{s}$ with the axis of rotation perpendicular to a magnetic field $B = 0.4\ \mathrm{T}$. If there are 100 turns in the loop, the maximum voltage induced in the loop is

(A) 15.07 V

(B) 150.7 V

(C) 1507 V

(D) 250 V

Q.8

Two identical induction coils each of inductance L joined in series are placed very close to each other such that the winding direction of one is exactly opposite to that of the other. What is the net inductance

(A) L^2

(B) $2\ \mathrm{L}$

(C) $L/2$

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(D) Zero

Q.9

A uniform flux density of $0.1\ \mathrm{Wb\ m^2}$, extends over a plane circuit of area $1\ \mathrm{m^2}$, and is normal to it. How quickly must the field be reduced to zero if an emf of $100\ \mathrm{V}$ is to be produced?

(A) $10^{-2}\mathrm{sec}$

(B) $10^{-3}\mathrm{sec}$

(C) $\frac{1}{10}\mathrm{sec}$

(D) 1 micro sec

Q.10

Self induction opposes the growth of current in a circuit when the current is

(A) switched on

(B) switched off

(C) steady

(D) fluctuating

Q.11

A current of 4 milli ampere is flowing through a coil having 200 turns. If the self inductance of the coil is 10 milli Henry then the magnetic flux through the cross section of the coil will be

(A) 2×10^{-7} Weber

(B) 3×10^{-7} Weber

(C) 1×10^{-7} Weber

(D) 3×10^{-6} Weber

Q.12

An infinitely long cylinder is kept parallel to a uniform magnetic field B directed along. positive z-axis. The direction of induced current as seen from the z axis will be

(A) Clockwise of the $+$ ve z axis

(B) Anticlockwise of the + ve z axis

(C) Zero

(D) Along the magnetic field

Q.13

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Two rails of a railway track insulated from each other and the ground are connected to a milli voltmeter. What is the reading of voltmeter, when a train travels with a speed of $180\ \mathrm{km/hr}$ along the track. Given that the vertical component of earth's magnetic field is 0.2×10^{-4} weber/m and the rails are separated by 1 metre.

(A) 10^{-2} volt

(B) 10^{-4} volt

(C) 10^{-3} volt

(D) 1 volt

Q.14

A coil of metal wire is kept stationary in a non-uniform magnetic field,

(A) an emf and current are both induced in the coil

(B) a current but no emf is induced in the coil

(C) an emf but no current is induced in the coil

(D) neither emf nor current is induced in the coil

Q.15

A loosely wound helix made of stiff wire is mounted vertically with the lower end just touching a dish of mercury. When a current from the battery is started in the coil through the mercury

(A) the wire oscillates

(B) the wire continues making contact

(C) the wire breaks contact just when the current is passed

(D) the mercury will expand by heating due to passage of current

Q.16

A small square loop of wire of side I is placed inside a large square loop of wire of side L $(L >> /)$. The loops are coplanar and their centres coincide. The mutual inductance of the system is proportional to

(A) $\frac{l}{L}$

(B) $\frac{l^2}{L}$

(C) $\frac{L}{l}$

(D) $\frac{L^2}{l}$

Q.17

A metallic wheel with 8 metallic spokes each of length r is rotating at an angular frequency ω in a plane perpendicular a magnetic field B. The magnitude of the induced emf between the axle and the rim of the wheel is

(A) $\frac{1}{2}\omega r^2 B$

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(B) $2\omega r^2 B$

(C) $4\omega r^2 B$

(D) $8\omega r^2 B$

Q.18

In an inductor, the current I (in ampere) varies with time t (in second)as $\mathrm{I} = 5 + 16\mathrm{t}$. If the emf induced in the inductor is $10\mathrm{mV}$, what is its self inductance ?

(A) $6.25 \times 10^{-4}\mathrm{H}$

(B) $6.25 \times 10^{-3}\mathrm{H}$

(C) $7.5 \times 10^{-4}\mathrm{H}$

(D) $7.5 \times 10^{-3}\mathrm{H}$

Q.19

A square loop of side I, mass m and resistance R falls vertically into a uniform magnetic field directed perpendicular to the plane of the coil. The height h through which the loop falls so that it attains terminal velocity on entering the region of magnetic field is given by

(A) $\frac{mgR}{2Bl}$

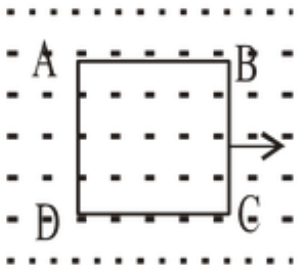
(B) $\frac{m^2gR^2}{2B^2l^2}$

(C) $\frac{mgR^2}{4B^3l^3}$

(D) $\frac{m^2gR^2}{2B^4l^4}$

Q.20

A metallic square loop ABCD is moving in its own plane with velocity u in a uniform magnetic field perpendicular to its plane as shown in Figure. An electric field is induced.



(A) in AD, but not in BC

(B) in BC, but not in AD

(C) neither in AD nor in BC

(D) in both AD and BC

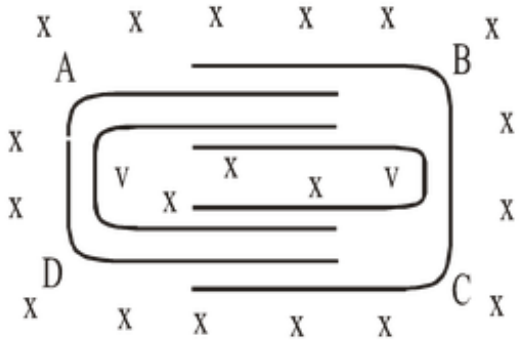
Q.21

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The ratio of secondary to primary turns is $9:4$. If power input is P, what will be the ratio of power output (neglect all losses) to power input

Q.22

One conducting U tube can slide inside another as shown in figure, maintaining electrical contacts between the tubes. The magnetic field B is perpendicular to the plane of the figure. If each tube moves towards the other at a constant speed V, then the emf induced in the circuit in terms of B, I and V where $/$ is the width of each tube, is $\mathrm{KB/N}$, where K is positive integer. Find K.



Q.23

The self inductance of the motor of an electric fan is $10\mathrm{H}$. In order to impart maximum power at $50\ \mathrm{Hz}$, it should be connected to a capacitance of K micro farad. Find K.

Q.24

When the current changes from $+2A$ to $-2A$ in 0.05 second, an e.m.f. of $8V$ is induced in a coil. The coefficient of self-induction of the coil is

Q.25

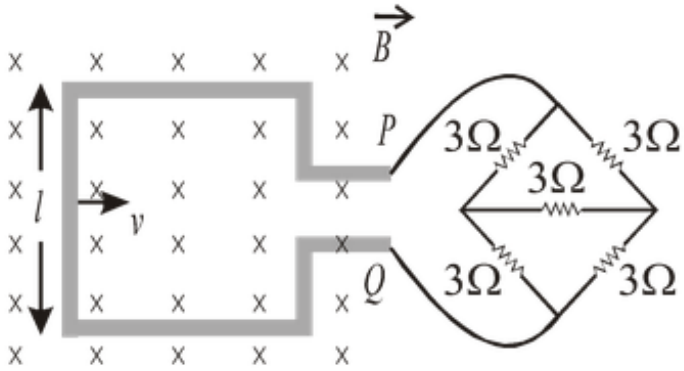
An aircraft with a wingspan of $40\ \mathrm{m}$ flies with a speed of $1080\ \mathrm{km\ h^{-1}}$ in the eastward direction at a constant altitude in the northern hemisphere, where the vertical component of earth's magnetic field is $1.75 \times 10^{-5}\ \mathrm{T}$. Then the e.m.f. that develops between the tips of the wings is e. Mark "10e" to nearest integer.

Q.26

A square metallic wire loop of side $0.1\ \mathrm{m}$ and resistance of 1Ω is moved with a constant velocity in a magnetic field of $2wb/m^2$ as shown in figure. The magnetic field is perpendicular to the plane of the loop, loop is connected to a network of resistance. What should be the velocity (in $\mathrm{cm/s}$) of loop so as to have a steady current

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of $1\ \mathrm{mA}$ in loop



Q.27

Flux ϕ (in weber) in a closed circuit of resistance 10Ω varies with time t (in seconds) according to the equation $\phi = 6\mathrm{t}^2 - 5\mathrm{t} + 1$. The magnitude of the induced current in the circuit at $t = 0.25\ \mathrm{s}$ is

Q.28

If a change in current of $0.01\ \mathrm{A}$ in one coil produces a change in magnetic flux of $2 \times 10^{-2}\ \mathrm{Wb}$ in the other coil, then the mutual inductance (in henry) of the two coils is

Q.29

A capacitor of capacitance $2\mu\mathrm{F}$ is charged to a potential difference of $12\ \mathrm{V}$. The charging battery is then removed and the capacitor is connected across an inductor of self inductance $0.6\mathrm{mH}$. The current in the circuit at a time when the potential difference across the capacitor is $6\ \mathrm{V}$ is

Q.30

In a transformer the primary has 500 turns and secondary has 50 turns. 100 volts is applied to the primary coil, the voltage developed in the secondary will be

Answers & Solutions

Q.1 Answer:

$\frac{\mathrm{i}_1}{\mathrm{i}_2} = \frac{1}{4}$

Solution:

From Faraday's Law, the induced voltage $\vee\alpha L$ rate of change of current is constant $\left(\mathrm{V} = -\mathrm{L}\frac{\mathrm{di}}{\mathrm{dt}}\right)$

$\therefore \frac{\mathrm{V}_2}{\mathrm{V}_1} = \frac{\mathrm{L}_2}{\mathrm{L}_1} = \frac{2}{8} = \frac{1}{4} \Rightarrow \frac{\mathrm{V}_1}{\mathrm{V}_2} = 4$

Power given to the two coils is same, i.e.,

$\mathrm{V}_1\mathrm{i}_1 = \mathrm{V}_2\mathrm{i}_2 \Rightarrow \frac{\mathrm{i}_1}{\mathrm{i}_2} = \frac{1}{4}$

Energy stored $\mathrm{W} = \frac{1}{4}\mathrm{Li}^2$

$\Rightarrow \frac{\mathrm{W}_2}{\mathrm{W}_1} = \left(\frac{\mathrm{L}_2}{\mathrm{L}_1}\right)\left(\frac{\mathrm{i}_2}{\mathrm{i}_1}\right)^2 = \left(\frac{1}{4}\right)(4)^2 = 4$

$\Rightarrow \frac{\mathrm{W}_1}{\mathrm{W}_2} = \frac{1}{4}$

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Q.2 Answer:

1 milli sec

Solution:

Peak current in the circuits $\mathrm{i_0} = \frac{12}{6} = 2\ \mathrm{A}$

Current decreases from $2A$ **to** $1A$**, i.e., becomes half in time**

$\mathrm{t} = 0.693\frac{\mathrm{L}}{\mathrm{R}} = 0.693 \times \frac{8.4\times10^{-3}}{6} = 1\mathrm{millisec}$

Q.3 Answer:

$0.1\ \mathrm{s}$

Solution:

$i = i_0 e^{-\frac{Rt}{L}}$

$\frac{1}{2}i_0 = i_0 e^{-\frac{2t}{30}}$

$\Rightarrow t = 0.1\mathrm{sec}$

Q.4 Answer:

$2.4\pi \times 10^{-4}\mathrm{H}$

Solution:

$\mathrm{M} = \frac{\mu_0\ \mathrm{N_1}\ \mathrm{N_2}\ \mathrm{A}}{1} = \frac{4\pi\times10^{-7}\times300\times400\times10\times10^{-4}}{0.2}$

$= 2.4\pi \times 10^{-4}\mathrm{H}$

Q.5 Answer:

$-10\ \mathrm{V}$

Solution:

$\frac{\mathrm{d}\phi}{\mathrm{dt}} = 20t - 50$

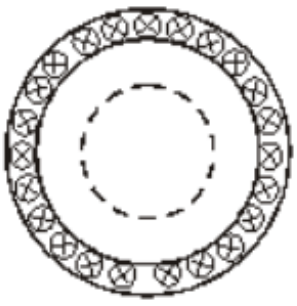
at $t = 3, \mathrm{emf} = -\frac{\mathrm{d}\phi}{\mathrm{dt}} = -10\ \mathrm{V}$

Q.6 Answer:

zero

Solution:

From ampere's law Crosssection of tube showing current and amperes loop



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$\int \mathrm{B} \cdot dl = \mu_0 I_{en}$

here $I_{en} = 0$ magnetic field at every point on the amper's loop is uniform due to symmetry.

$\therefore \mathrm{B} = 0$ **at every point inside the tube.**

Q.7 Answer:

1507 V

Solution:

$V_{\max} = BANw = 0.4 \times 0.1 \times 100 \times 2\pi 60 = 1507.2$

Q.8 Answer:

Zero

Solution:

When the two coils are joined in series such that the winding of one is opposite to the other, then the emf produced in first coil is 180° out of phase of the emf produced in second coil.

Thus, emf produced in first coil is negative and the emf produced in second coil is positive so, net inductance is

$\mathrm{L} = \mathrm{L}_1 + \mathrm{L}_2 = \mathrm{L} + \mathrm{L} \Rightarrow \mathrm{L} = \frac{\phi}{\mathrm{i}} + \frac{\phi}{\mathrm{i}} = 0$

Q.9 Answer:

10^{-3}sec

Solution:

$\phi_1 = \mathrm{BA} = 0.1 \times 1 = 0.1 \text{ Wb}; \phi_2 = 0$

$\mathrm{e} = -\frac{\mathrm{d}\phi}{\mathrm{dt}} = \frac{\phi_2 - \phi_1}{\mathrm{t}}$

$100 = -\left(\frac{0-0.1}{\mathrm{t}}\right), \mathrm{t} = 10^{-3} \text{ s}$

Q.10 Answer:

switched on

Solution:

When the current is switched on, e.m.f. is induced in the circuit and this further opposes the growth of current in the circuit.

Q.11 Answer:

2×10^{-7} Weber

Solution:

$\phi = \mathrm{LI} = 10 \times 10^{-3} \times 4 \times 10^{-3} = 4 \times 10^{-5} \text{ Wb}$

Magnetic flux through cross section of the coil

$= \frac{\phi}{\mathrm{n}} = \frac{4 \times 10^{-5}}{200} = 2 \times 10^{-7} \text{ Wb}$

Q.12 Answer:

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Zero

Solution:

Since the magnetic field is uniform therefore there will be no change in flux hence no current will be induced.

Q.13 Answer:

10^{-3} volt

Solution:

$E = B_v \cdot vl$
$= 0.2 \times 10^{-4} \times \left(\frac{180 \times 1000}{3600}\right) \times 1 = 10^{-3}\ \mathrm{V}$

Q.14 Answer:

neither emf nor current is induced in the coil

Solution:

If a coil is not moved in a magnetic field, the magnetic flux does not change. Hence no emf or current is induced in the coil.

Q.15 Answer:

the wire oscillates

Solution:

When a current is passed through the helix, the neighbouring coils of the helix attract each other due to which it contracts. As a result the contact is broken and the coils will recover their original state under the influence of a restoring force. The contact is made again and the process continues. Thus the wire oscillates.

Q.16 Answer:

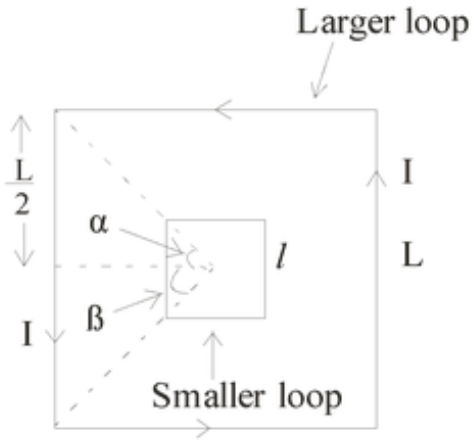
$\frac{l^2}{L}$

Solution:

The magnetic field due a current I in the large loop at its centre is $\mathrm{B} = 4$ **times that due to one side**
$= 4 \times \frac{\mu_0}{4\pi} \frac{\mathrm{I}}{(\mathrm{L}/2)} (\cos\alpha + \cos\beta) = \frac{2\mu_0 \mathrm{I}}{\pi \mathrm{L}} (\cos 45^\circ + \cos 45^\circ)$
$= \frac{2\sqrt{2}\mu_0 \mathrm{I}}{\pi \mathrm{L}} (\because \alpha = \beta = 45^\circ)$
The magnetic flux that links the larger loop with the smaller loop of side $\mid (I << L)$ **is**

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$\phi_{12} = \mathrm{Bt}^2 = \frac{2\sqrt{2}\mu_o \mathrm{I}^2}{\pi \mathrm{L}}$



∴ Mutual inductance

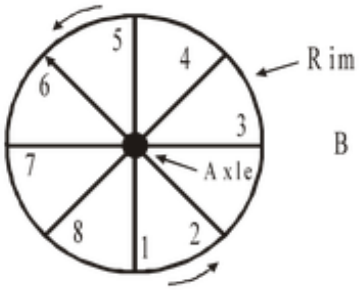
$\mathrm{M}_{12} = \frac{\phi_{12}}{\mathrm{I}} = \frac{2\sqrt{2}\mu_0}{\pi}\left(\frac{\mathrm{l}^2}{\mathrm{L}}\right)$

i.e. $\mathrm{M}_{12} \propto \frac{\mathrm{l}^2}{\mathrm{L}}$

Q.17 Answer:

$\frac{1}{2}\omega r^2 B$

Solution:



Refer to Fig. Let v be the frequency of rotation. The time taken for 1 full rotation is $\mathrm{T} = 1/\mathrm{v}$

Therefore, rate of change of area is

$\frac{\mathrm{A}}{\mathrm{T}} = \frac{\pi \mathrm{r}^2}{\mathrm{T}} = \pi \mathrm{r}^2 \mathrm{v}$

Now, the emf induced between the axle and rim is $\mathrm{e} = \mathrm{B} \times$ **rate of change of area**

$= \mathrm{B} \times \pi \mathrm{r}^2 \mathrm{v} = \frac{1}{2}\mathrm{Br}^2\,\omega (\because \omega = 2\pi \mathrm{v})$

Since the same emf is produced between the ends of each spoke, and these emfs are in parallel as is evident from Fig. the net emf between the axe and the rim of the wheel will be the same as that across each spoke. We notice that all the eight spokes are connected with one end at the rim and the other at the axle. Hence the magnitude of the net emf between the axle and the rim is independent of the number of spokes.

Q.18 Answer:

$6.25 \times 10^{-4}\mathrm{H}$

Solution:

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Now $|\mathrm{e}|=\mathrm{L}\frac{\mathrm{dI}}{\mathrm{dt}}$
or $10\times10^{-3}=\mathrm{L}\times\frac{\mathrm{d}}{\mathrm{dt}}(5+16\mathrm{t})=\mathrm{L}\times(16)$
or $L=6.25\times10^{-4}\mathrm{H}$

Q.19 Answer:

$\frac{m^2gR^2}{2B^4l^4}$

Solution:

Velocity $u=\sqrt{2gh}$**. Induced emf** $\mathrm{e}=\mathrm{Blu}=Bl\sqrt{2gh}$**. Therefore, the induced current in the loop is**
$I=\frac{Bl\sqrt{2gh}}{R}$
$\therefore$ Force $F=BIl=\frac{B^2l^2\sqrt{2gh}}{R}$
The loop will attain terminal velocity if this force equals mg**, i.e. if**
$\frac{B^2l^2\sqrt{2gh}}{R}=mg$
which gives $h\frac{m^2gR^2}{2B^4l^4}$

Q.20 Answer:

in both AD and BC

Solution:

Electric field will be induced in both AD **and** BC**, since both are moving perpendicular to the direction of the magnetic field and the flux linked with them is changing with time.**

Q.21 Answer:

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Solution:

Since all the losses are neglected.
So $P_{\text{out}}=P_{\text{in}}$

Q.22 Answer:

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Solution:

Induction is due to change in area.
$\frac{\mathrm{d}\phi}{\mathrm{dt}}=\mathrm{B}\frac{dA}{dt}$
$=\mathrm{B}.2.I\ \mathrm{V}(\text{Because relative velocity }=2\ \mathrm{V})$

Q.23 Answer:

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Solution:

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For maximum power: $\omega = \frac{1}{\sqrt{LC}}$

$\Rightarrow 2p \times 50 = \frac{1}{(10 \times C)^{1/2}}$

$\Rightarrow C = 1\mu_F$

Q.24 Answer:

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Solution:

$\mathrm{e} = \mathrm{L}\frac{\mathrm{di}}{\mathrm{dt}} \Rightarrow 8 = \mathrm{L} \times \frac{(2-(-2))}{0.05} \Rightarrow \mathrm{L} = 0.1\mathrm{H}$

Q.25 Answer:

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Solution:

$L = 40\mathrm{mv} = 1080 \text{ km h}^{-1} = 300 \text{ msec}^{-1}$

$\mathrm{B} = 1.75 \times 10^{-5} \text{ T}$

$\therefore \mathrm{e} = \mathrm{Blv} = 1.75 \times 10^{-5} \times 40 \times 300 = 0.21 \text{ V}$

Q.26 Answer:

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Solution:

Equivalent resistance of the given Wheatstone bridge circuit (balanced) is 3Ω **so total resistance in circuit is** $\mathrm{R} = 3 + 1 = 4\Omega$**. The emf induced in the loop** $e = Bvl$**.**

So induced current $i = \frac{e}{R} = \frac{Bvl}{R}$

$\Rightarrow 10^{-3} = \frac{2 \times v \times \left(10 \times 10^{-2}\right)}{4} \Rightarrow v = 2 \text{ cm/sec}$

Q.27 Answer:

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Solution:

The rate of change of flux gives the induced emf.

Thus $\mathrm{e} = -\frac{\mathrm{d}\phi}{\mathrm{dt}} = -\frac{\mathrm{d}}{\mathrm{dt}}\left(6\mathrm{t}^2 - 5\mathrm{t} + 1\right)$

$= -12t + 5$

At $t = 0.25 \text{ s}, \mathrm{e} = -12 \times 0.25 + 5 = -3 + 5 = 2 \text{ V}$

$\therefore$ **Induced current** $\mathrm{I} = \frac{\mathrm{e}}{\mathrm{R}} = \frac{2}{10} = 0.2 \text{ A}$

Final ans is 0.2 **A.**

Q.28 Answer:

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Solution:

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$\mathrm{M}=\frac{\Delta \phi}{\Delta \mathrm{t}}=\frac{2 \times 10^{-2}}{0.01}=2$ **henry**
$i_1=0.01 \mathrm{~A} ; \phi_2=2 \times 10^{-2} \mathrm{wb}$
but $\phi_2=\mathrm{M} i_1$

$$\Rightarrow \mathrm{M}=\frac{\phi_2}{i_1}=\frac{2 \times 10^{-2}}{0.01}=2$$

Q.29 Answer:

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Solution:

We know that $\mathrm{Q}=\mathrm{CV}$ **and** $\mathrm{Q}=\mathrm{Q}_0 \cos \omega \mathrm{t}$. **Also** $\mathrm{Q}_0=\mathrm{CV}_0$
$\therefore \cos \omega \mathrm{t}=\frac{\mathrm{Q}}{\mathrm{Q}_0}=\frac{\mathrm{V}}{\mathrm{V}_0}=\frac{6}{12}=\frac{1}{2}$ **or** $\omega \mathrm{t}=\frac{\pi}{3}$
Now ω **is given by** $\omega=\frac{1}{\sqrt{\mathrm{LC}}}$
Given $\mathrm{L}=0.6 \times 10^{-3} \mathrm{H}$ **and** $\mathrm{C}=2 \times 10^{-6}$ F. **Using these values in Eq. (1)**
we get $\omega=\frac{10^5}{2 \sqrt{3}} \mathrm{rads}^{-1}$
Now $\mathrm{I}=\frac{\mathrm{dQ}}{\mathrm{dt}}=\frac{\mathrm{d}}{\mathrm{dt}}\left(\mathrm{Q}_0 \cos \omega \mathrm{t}\right)=-\mathrm{Q}_0 \omega \sin \omega \mathrm{t}$
$\therefore|\|=Q_0 \omega \sin \omega t=C V_0 \omega \sin \omega t=\left(2 \times 10^{-6}\right) \times 12 \times \frac{10^5}{2 \sqrt{3}} \sin \left(\frac{\pi}{3}\right)=0.6 \mathrm{~A}$

Q.30 Answer:

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Solution:

$\frac{N_s}{N_p}=\frac{V_s}{V p}$
$\frac{50}{500}=\frac{V_s}{100}$
$V_s=10 \mathrm{~V}$

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